MIE 446

Aerospace Structures Fall 2026 Syllabus

Revised September 26 2026

Course information	Fall 2026
Credits / mode	3 credits • In person
Meetings	Tuesday & Thursday, 11:30 a.m.–12:45 p.m.
Classroom	Engineering Laboratory (ELAB), Room 323
Instructor	Dr. Ehsan Roohi Golkhatmi • roohie@umass.edu
Office	Gunness Laboratory, Room 1 • Office hours: Fri. 12:20-1:10 PM (Other hours by appointment)
Prerequisite	MIE 313 — Design of Mechanical Components
Course site	Canvas is the authoritative source for announcements and submissions

Course at a glance

Aerospace structures are light because they must fly, yet they must safely carry bending, shear, torsion, landing, and maneuver loads. This course connects those loads to mechanical models and to a buildable wing. The opening bridge introduces airfoils and finite wings for students who have not taken an aerospace course; the remainder develops the structural tools needed to analyze a wing and the design-for-manufacture skills needed to code, print, inspect, and communicate a physical build.

Signature experience

Teams of three will use instructor-provided code to generate a small parametric wing, verify its geometry, export printable components, and produce a documented physical build. After assembly, every team will complete an unloaded, non-contact inspection with the shared Jetson-camera station and analyze a supervised structural-health-monitoring demonstration performed on a separate sacrificial specimen. Teams will be evaluated through reproducible configuration and verification, calibrated manual/vision measurements, annotated images and data logs, print and assembly quality, interpretation of before/after damage evidence, uncertainty, revision history, and technical defense.

Central question: What makes an engineering answer trustworthy?

> **Course commitment. AI can generate equations, code, plots, and polished explanations, but a correct-looking answer is not enough. Students will learn to frame the physical problem, trace the load path, predict behavior, select and justify a model, audit human/computational/AI results, interpret evidence, and defend a safe decision. Analytical work builds the internal physical model needed to supervise tools. Students - not tools - own the judgment, teamwork, physical evidence, and consequences.**

1. Purpose, scope, and learning outcomes

Course description

The course surveys classical structural mechanics in an aerospace setting. Topics include aircraft forces and airfoil/wing geometry; internal loads and load paths; stress and strain in two and three dimensions; elasticity assumptions; beam bending and deflection; energy methods and virtual work; aircraft maneuver and landing loads; V-n diagrams; structural idealization; and shear flow, torsion, and shear center in thin-walled wing structures. The course follows a physics-centered approach: computational or AI output is treated as a claim to be validated against governing physics, transparent analytical baselines, known cases, and physical evidence.

Learning outcomes

- Trace aerodynamic, inertial, and support loads through an aerospace structure and construct defensible free-body diagrams and load paths.
- Predict signs, trends, deformation shapes, relative magnitudes, critical locations, and likely failure modes before detailed calculation.
- Select and justify analytical idealizations; state assumptions, governing relationships, boundary conditions, and limits of validity.
- Compute stress transformations, principal stresses, beam stresses and deflections, strain energy, and selected redundant reactions.
- Construct and interpret shear-force, bending-moment, torque, and V-n diagrams; analyze idealized and thin-walled wing sections.
- Use and explain instructor-provided parametric and measurement code, apply additive-manufacturing constraints, fabricate a wing, and document dimensional compliance, interface fit, machine-vision inspection, print quality, and assembly readiness.
- Audit human-, computational-, and AI-generated results using dimensions, equilibrium, boundary and limiting cases, independent estimates, sensitivity, and physical evidence; diagnose plausible errors and their consequences.
- Translate results, uncertainty, and model limitations into a defensible engineering decision and transfer the reasoning to an unfamiliar case.
- Communicate and defend reasoning clearly; collaborate with empathy and professional responsibility; seek feedback, revise, and identify what must be learned next.

Learning sequence

Phase	What students learn	Evidence
Aero bridge	NACA exercise; prediction-before-calculation; checked load map	NACA coding exercise; load map
Mechanics core	HW 1-5 judgment records; worked/AI-solution audit; midterm	HW 1–5; midterm
Wing structures	HW 6; analytical baseline; validated code; design reviews	HW 6; design reviews
Build–inspect–learn	Parametric geometry, slicing, fabrication, calibrated manual/machine-vision inspection, and shared damage diagnostics	Configured code, wing, print log, annotated inspection image/CSV, diagnostic analysis, report, presentation

Human capabilities in an AI-rich world

Class time is intentionally organized around doing, explaining, questioning, building, and reflecting. The following capabilities are explicit course outcomes, not incidental benefits:

Teamwork: Three-person teams rotate roles, review work, resolve interfaces, and share accountability.

Hands-on experience: Students fabricate, inspect, and document a physical wing instead of stopping at a digital answer.

Creativity: Teams generate and compare design alternatives within real load, mass, material, and printer constraints.

Empathy: Students listen, share work fairly, account for access and safety needs, and recognize how engineering choices affect people.

Human communication: Briefs, oral checks, peer critique, guest interaction, and the final defense require clear, respectful dialogue.

Critical thinking: Students challenge assumptions, verify code and AI output, compare manual and machine-vision measurements, interpret force/deflection, vibration, and crack evidence, and reconcile analytical expectations with geometry, slicer evidence, and the completed build.

Lifelong learning: Students document debugging, seek feedback, revise after failure, and identify what they need to learn next.

2. People, communication, and resources

Role	Name	Email	Availability
Instructor	Dr. Ehsan Roohi Golkhatmi	roohie@umass.edu	Gunness 1; Fri. 12:20-1:10 PM
Undergraduate teaching assistant	Ava Towfigh	atowfigh@umass.edu	TBA
Undergraduate teaching assistant	Lev Kudriavtsev	lkudriavtsev@umass.edu	TBA

Technical questions are best handled in class or office hours. Send grading and accommodation questions to the instructor, not to the full class. Allow two business days for email responses.

Guest learning events

Date	Speaker	Plan
Fri., Sept. 25(outside class)	Dr. Avshalom Manela, Technion, Associate Professor and Vice Dean for Teaching; research in rarefied-gas dynamics, small-scale flows, heat/mass transfer, aeroacoustics, and fluid–structure interaction.	Optional research seminar: final time, title, and room will be announced in Canvas. Pizza lunch afterward. Attendance/reflection may earn the published extra credit.
Thu., Oct. 2911:30–12:45ELAB 323	Dr. Michael Tsay, Vice President of Technology, Busek Co. Inc.	Intro to Electric Propulsion: Hall-effect thrusters and gridded ion engines; industry practice and pathways to internships/careers at Busek. This meeting replaces the regular lecture.

Speaker background: Dr. Manela's Technion profile. Dr. Tsay will also discuss industry practice and pathways to internships and careers at Busek.

Optional seminar extra credit

The Sept. 25 event is outside class and optional. Attendance plus a 150-word technical reflection submitted within 48 hours earns up to 2.5 course percentage point. Students unable to attend may complete the same reflection using another aerospace/industry seminar; this equal-access alternative earns identical credit.

3. Assessment and evidence of learning

Component	Basis	Course weight
Homeworks (1–6)+ Midterm examination (counts as two HW)	8× 8.75%	70%
Team wing fabrication, inspection, diagnostics, report, and presentation	Milestones, instructor-provided code verification, fabrication, calibrated inspection, shared damage-data analysis, report, and presentation	30.0%
Total		100%

There is one in-class midterm (Tuesday, Oct. 27) and no final examination.

Homework portfolio: six assignments

Unless a handout specifies otherwise, each homework submission is individual and contains the following artifacts. HW 3 requires only one PDF with analytical solutions and software comparisons:

1. Analytical solution.
2. One-slide Claim-Evidence-Check brief (.pptx and exported .pdf): briefly describe the problems you solved in that HW. The slide is a synthesis, not a screenshot collage.
3. Process and AI note: Explain how you used AI in answering the HW.

Midterm

The midterm on October 27 is individual and closed to generative AI and unauthorized online assistance but open to other permitted resources. It covers material taught through October 22; the detailed scope and practice problems will be posted in Canvas.

Letter grades

Grade	%	Grade	%	Grade	%
A	93–100	B−	80–82.9	D+	67–69.9
A−	90–92.9	C+	77–79.9	D	63–66.9
B+	87–89.9	C	73–76.9	F	<63
B	83–86.9	C−	70–72.9		

UMass undergraduate grading does not use A+ or D−. Any end-of-term adjustment will be uniform and will never lower a student's earned letter grade.

4. Project brief: Code-to-Print Wing - Fabrication, Machine-Vision Inspection, and Damage Diagnostics

Design intentBuild and verify one code-generated, non-flying wing; inspect the completed team wing without contact and without load using calibrated machine vision; and interpret a shared before/after-damage demonstration on a separate sacrificial wing-box specimen. Success requires traceable geometry, manufacturing evidence, measurement agreement and uncertainty, physically plausible diagnostic trends, and a defensible engineering decision.

Assigned project teams

Three-person teams have already been assigned. The wing project begins in class on September 29. Roles rotate among geometry and verification, CAD and design for additive manufacturing, fabrication and documentation, and measurement and data interpretation. Every member must understand the complete workflow and answer individual technical questions.

Common design envelope

Constraint	Baseline requirement
Artifact	One code-generated, non-flying wing demonstrator. Each team wing remains unloaded and undamaged; no propulsion, onboard electronics, controls, fuselage, flight activity, or force-to-failure test
Geometry	One 450 mm semi-wing; 160 mm root chord; 100 mm tip chord; an instructor-approved NACA four-digit section. The full-wing area and aspect ratio are analytical mirrored-wing equivalents.
Construction	Two or three printed modules; shell and ribs; two course-issued 4 mm rods at x/c = 0.30 and 0.60 in the released guided notebook. Any departure requires instructor approval before print release.
Material budget	Up to 300 g regular PLA Pro per final wing, plus course-issued rods and supervised 5-minute epoxy when assigned
Code	Instructor-provided Python/CadQuery wing-generation code plus supervised Python/OpenCV and data-acquisition workflows; teams configure inputs, run cleanly, verify outputs, explain the logic, and preserve a reproducible record
Engineering checks	Units; planform metrics; bounding boxes; watertight solids; printable thickness; build-envelope/mass checks; camera calibration; manual-to-vision agreement; load-cell zero/sanity check; accelerometer sampling and frequency sanity check
Build acceptance	Complete labeled modules; approved dimensions; dry-fit interfaces; aligned assembly; no major cracks, warping, separation, or missing features; manual and Jetson-camera evidence; complete archive
Inspection and diagnostics	Required unloaded Jetson-camera inspection for every team wing. Shared instructor-led demonstration uses force, camera tip-deflection, acceleration, and microscope evidence on a sacrificial specimen before and after controlled damage.
Evaluation	Understanding, reproducibility, traceability, print quality, fit, calibrated inspection, interpretation of diagnostic evidence, uncertainty, revision, communication, and individual defense - not a beauty contest

The instructor provides the guided Python/CadQuery wing-generation notebook and the Jetson/OpenCV/data-acquisition workflows. Teams configure, run, verify, explain, and defend these tools; writing a CAD kernel from scratch is not required. AI use remains permitted with disclosure and an independent check.

Fabrication starts on Tuesday, October 13, in assigned, supervised ELab slots. A team may release its first wing module only after a physical rod-fit coupon, a documented calculation check, a clean notebook run, a slicer review, and staff print approval. October 13 is a production start, not a promise that all 21 teams will finish printing that day.

Project grading (30% of the course)

Element	Evidence	Weight
Milestones and design reviews	Requirements, concept, role evidence, inspection plan, calibration plan, and risk controls	5%
Reproducible workflow and design rationale	Correct parameters, geometry, instructor-provided code configuration, verification checks, files, and README	8%
Fabrication, inspection, and build acceptance	Safe process, print/assembly quality, manual and calibrated machine-vision evidence, annotated image/CSV, and evidence archive	8%
Final technical report	Concise technical argument integrating geometry, manufacturing, inspection uncertainty, and before/after force-deflection, vibration, and crack evidence	5%
Presentation and individual defense	Team story, clear visuals, diagnostic interpretation, questions, and peer evidence	4%

5. Project workflow and deliverables

Target date	Deliverable or gate
Sept. 22	Three-person teams already assigned; project roles and shared folders confirmed.
Sept. 29-Oct. 1	Project launch and manual geometry, planform, and load-path calculations.
Oct. 6-8	Supervised coupon-only run, physical fit test, rod measurement, and R01 record. Staff schedules the short coupon appointments.
Oct. 8-9	Pre-print review: calculation sheet, coupon evidence, R02 clean run, module dimensions, CAD mass, all-layer slicer review, safety training, and assigned printer slot.
Oct. 13	Supervised wing-module fabrication begins for released teams. Teams without a passed coupon or release work on the missing gate, not a final wing print.
Oct. 22	Reproducibility and as-built checkpoint: parameter trace, file/revision consistency, print log, independent check, and current risks.
Oct. 26; Nov. 2	Follow-up ELab slots for remaining modules, authorized corrections or reprints, and dimensional checks; reservations and machine capacity are posted in Canvas.
Nov. 12; Nov. 16	Design and assembly review, dry fit, alignment plan, supervised assembly or authorized correction, and early manual inspection.
Dec. 1-3	Build acceptance and calibrated manual/Jetson inspection; shared damage-diagnostics demonstration and data analysis.
Dec. 8, 10, 15	Team presentations and individual technical questions; final package due Dec. 15 at 5:00 p.m.

Submission package

- Instructor-provided source code/notebooks plus the team's configuration, environment file, README, and clean runs that reproduce geometry, plots, inspection output, and assigned diagnostic analysis.
- STL/3MF files, slicer project, G-code only when requested, every-layer screenshots, print log, calibration target image, manual measurements, one annotated vision image, and concise CSV/log.
- Design, Fabrication, and Inspection Memo with requirements, parameters, code checks, dimensional and slicer evidence, manual-to-vision comparison, uncertainty, limitations, build-acceptance results, diagnostic interpretation, and justified revisions.
- Completed unloaded wing, signed build-acceptance checklist, print log, first-layer and completed-part photographs, actual mass, dry-fit and alignment evidence, and revision record for any failed attempt.
- Final report and concise team presentation that connect mechanics predictions, manufacturing evidence, calibrated inspection, and shared before/after damage evidence; every member answers an individual technical question.

Required post-build inspection and diagnostic activities

1. **Team-wing inspection (required). Place the unloaded assembled wing in the alignment jig with ruler and ArUco/checkerboard references. Use the shared Jetson Orin Nano, Logitech webcam, camera stand, and instructor-provided Python/OpenCV workflow to estimate semi-span, root/tip chord, seams, and gross alignment. Compare against ruler and digital-caliper measurements and submit calibration evidence, one annotated image, a CSV/log, differences, uncertainty, and at least one identified failure mode.**
2. **Shared structural-health-monitoring demonstration (required analysis). Using a separate sacrificial 3D-printed wing-box specimen, the instructor/TA will acquire force with the 10 kg load cell and HX711, camera-based tip deflection, ADXL343 acceleration, and USB-microscope crack images before and after controlled damage. Teams will compute or interpret stiffness from the force-deflection slope, dominant natural frequency from the acceleration record, and qualitative crack change; then decide which indicators provide credible evidence of damage and which do not.**
3. **Safety and specimen boundary. No submitted team wing will be loaded, cracked, or tested to failure. Controlled damage is introduced only by the instructor on designated sacrificial specimens; loading is slow, clamped, shielded with clear polycarbonate, and performed with required eye protection. Students do not alter the fixture, damage a specimen, or enter the protected loading area without explicit authorization. A failed print or measurement disagreement can support learning when the evidence, likely cause, and one controlled correction are documented; hiding a result or bypassing a design or safety gate is a violation.**

Recommended team operating rhythm

Every week	Evidence
Plan	One measurable team goal; owner and due date for each action

Every week	Evidence
Integrate	Code and geometry reviewed by a teammate who did not create them
Verify and reflect	One independent physics/manual check; one calibration or sensor sanity check when applicable; one risk updated in the design log; brief contribution record and decision rationale

6. ELab fabrication requirements

Location and access
Printer room: Engineering Laboratory (ELab), Room 104 / 104A.

Approved equipment and material

Use the Creality K2 Pro + CFS or an Ender Pro only as assigned by ELab staff, with the exact printer and 0.4 mm nozzle profile. The baseline course material is regular 1.75 mm PLA Pro. Do not substitute Overture Air PLA, Aero PLA, or another lightweight/foaming filament: last year's course experienced brittleness, poor bed adhesion, and failed parts. If a lightweight filament is separately approved, feed it from the external spool rather than the CFS.

Before the slot

- Complete the agreement/orientation; work only during the assigned team slot and with required supervision.
- Bring frozen watertight STL/3MF files, the project's print-readiness approval, and a named team operator.
- Confirm printer, nozzle, filament profile, part dimensions, orientation, supports, wall/rib thickness, spar sockets, estimated mass, and estimated time.
- Slice and inspect every layer—not only the outside preview. Save the slicer project (.3mf) before exporting G-code.

At the printer

- Check that the bed is clean, the correct filament is dry/loaded, and the build plate is seated. Use adhesive only as trained.
- Start the job under supervision and remain for the first layers. Confirm adhesion, extrusion, walls, and supports before leaving the machine in the approved monitored state.
- Never operate alone; never bypass alarms or guards; no food or drink in the room; do not remove another team's job or change machine settings without permission.
- Record printer, material, slicer settings, start/end times, mass, failures, and corrective actions in the print log.

After printing and assembly

- Let parts cool, remove them as trained, clean the station, label all parts, and weigh the final printed set.
- Dry-fit rods and modules before epoxy. Sand only as authorized; use gloves, mixing tools, ventilation, and the instructor's fixture/jig.
- Use 5-minute epoxy only when assigned and under supervision; clamp the assembly in alignment and follow the full handling and cure guidance before moving it.
- Report a failure promptly. Do not start an unscheduled reprint; the instructor/TA will approve the smallest useful revision after reviewing the print log and photographs.

Course-provided project materials

Course-provided item	Student responsibility
Regular 1.75 mm PLA Pro	Use the assigned material/profile and remain within the approved project mass budget
Course-issued carbon-fiber rods/interfaces	Use the dimensions posted in Canvas and verify dry fit before any adhesive
5-minute epoxy and shared consumables	Use only when assigned, under supervision, and with required lab controls
Shared inspection and diagnostic station	Use only during assigned supervised sessions: Jetson Orin Nano, webcam/stand, ruler/calibration target, digital caliper, load cell/HX711, ADXL343 accelerometer, breadboard/jumpers, USB microscope, clamps, clear shield, and safety glasses
Personal purchases or substitutions	Not required; obtain written approval before bringing another material, chemical, or tool

The course supplies approved PLA Pro, course-issued rods/interfaces, supervised adhesive, shared fabrication consumables, and one shared inspection/diagnostic station. The instructor provides the baseline wing-generation, machine-vision, and data-acquisition code. Students should not purchase or substitute structural materials, electronics, chemicals, or tools unless the instructor gives written approval. The complete machine workflow, inspection instructions, and acceptance checklist are provided on Canvas.

7. Weekly schedule — Fall 2026

The four 75-minute meetings before fabrication are specified below. October 12 is a University holiday; there is no regular class that day. Coupon appointments outside class require ELab authorization and supervision. The revised due dates below supersede the earlier syllabus schedule; Canvas provides submission links and team-specific printer reservations. No team may bypass a safety or print-release condition because of the accelerated calendar.

Class plan before October fabrication

Meetings run from 11:30 a.m. to 12:45 p.m. The remaining part of Lecture 1 is completed on September 29. Its fuselage and monocoque extensions become reading rather than delaying the wing calculation and print-release work.

Meeting	In-class sequence and time	Evidence at end of meeting
Tue. Sept. 29	11:30-11:40 retrieve tail-to-wing vocabulary; 11:40-12:05 finish Lecture 1 wing parts and devices (7A-7A-4), including a brief spoileron/wing-tip-rudder distinction; 12:05-12:25 trace skin-rib-rod/root load paths and the 7C wing cut; 12:25-12:40 map those parts to the team semi-wing; 12:40-12:45 exit check.	Labeled half-wing, root-to-tip load path, and one unanswered design question.
Thu. Oct. 1	11:30-11:40 retrieval and units; 11:40-12:05 semi-span, chord law, taper, projected semi-wing area, mirrored full-wing area and aspect ratio; 12:05-12:25 hand calculation for the 450/160/100 mm baseline; 12:25-12:40 independent peer check against the guided notebook; 12:40-12:45 exit check.	Signed geometry sheet with units and a separately checked result. HW 1 due.
Tue. Oct. 6	11:30-11:40 decode an approved NACA four-digit section; 11:40-12:00 place the fixed rods, ribs and module cuts and check build-envelope and wall limits; 12:00-12:20 distinguish radial from diametral clearance and predict coupon-hole sizes; 12:20-12:40 run Coupon Only and plan the supervised physical fit test; 12:40-12:45 exit check.	Coupon R01 archive and booked staff-supervised coupon test; measured fit record follows the physical print.
Thu. Oct. 8	11:30-11:45 draw the cantilever free-body diagram for one instructor-specified virtual load (no load is applied to the student wing); 11:45-12:05 check shear/root moment, module dimensions and CAD/slicer mass; 12:05-12:35 peer review coupon evidence, R02 clean run, every-layer preview and revision trace; 12:35-12:45 staff release or written correction list.	Pre-print release packet or explicit hold. No final-wing print without a physical coupon and staff authorization.

Required before the first wing module: semi-wing and full-wing geometry with units; rod and module locations; a one-half-wing free-body diagram and independently checked shear/root-moment demand for the assigned virtual load; measured rod diameter and physical coupon fit; a clean final-wing R02 run; CAD mass at or below 300 g; checked module bounds and every slicer layer; safety authorization and a team-specific printer reservation. A missing coupon or failed check keeps the team in the coupon/prototype workflow.

Week / dates	Topics and learning activities	Due / event
1Sept. 8/10	Central question: what makes an answer trustworthy? Why aircraft structures are different; four forces; airfoil geometry, angle of attack, pressure, lift/drag/moment; NACA 4-digit coding studio; prediction-before-calculation routine	Starter environment check
2Sept. 15/17	Finite-wing geometry, skin-rib-spar load paths, and cantilever idealization; existing three-person teams prepare for the project launch.	Teams already assigned; no new team-formation task.
3Sept. 22/24	External and internal loads, equilibrium, stress and strain, and wing free-body diagrams.	Existing teams active; Fri. Sept. 25 optional Manela seminar.
4Sept. 29/Oct. 1	Sept. 29: finish Lecture 1 wing anatomy, devices, spoileron comparison, and structural load paths. Oct. 1: semi-wing geometry, taper, area, aspect ratio, and independent hand check.	HW 1 due Oct. 1; team calculation sheet started.
5Oct. 6/8	Oct. 6: NACA section, rods, modules, printer constraints, and physical fit coupon. Oct. 8: cantilever free-body diagram, virtual-load shear and root moment, peer audit, and print-release review.	Coupon R01 and fit record by Oct. 8; pre-print package reviewed Oct. 8-9.
6Oct. 13/15	Oct. 13: supervised wing-module fabrication launch in assigned slots, with a beam-equilibrium studio for teams waiting their turn. Oct. 15: beam shear/moment, bending stress, slope and deflection; inspect first-print evidence.	Wing construction starts Oct. 13 only for released teams; HW 2A (Parts 1 and 3, 35 points) due Oct. 20.
7Oct. 20/22	Two- and three-dimensional stress transformation; Mohr's circle, principal stresses and directions; constitutive and plane-stress assumptions. Audit one plausible but flawed solution.	As-built and reproducibility checkpoint Oct. 22.
8Oct. 27/29	Midterm Oct. 27; guest class Oct. 29 on electric propulsion, industry practice, and internship/career pathways.	Midterm Oct. 27; Dr. Tsay Oct. 29.
9Nov. 3/5	Nov. 3: no class (University schedule). Nov. 5: strain energy, virtual work, and a beam-model validity check.	HW 3 (three beam cases, 65 points) due Nov. 5.

Week / dates	Topics and learning activities	Due / event
10Nov. 10/12	Nov. 10: Castigliano and determinate/indeterminate structures. Nov. 12: maneuver and landing loads, V-n diagrams, wing structural idealization, and assembly review.	Design and assembly review.
11Nov. 17/19	Thin-walled wing-box idealization, shear flow, torsion and shear center; compare the structural model with printed geometry and inspection evidence.	HW 4 due Nov. 19; Nov. 16 supervised assembly/inspection slot.
12Nov. 24/26	No regular MIE 446 meetings: Nov. 24 follows a Wednesday schedule; Thanksgiving recess.	Approved inspection or correction slots only if the shared schedule permits.
13Dec. 1/3	Manual and calibrated Jetson machine-vision inspection, uncertainty, build acceptance, and shared force-deflection, vibration, and crack demonstration.	HW 5 due Dec. 3; inspection and diagnostic evidence.
14Dec. 8/10	Build acceptance, diagnostic synthesis, team presentations, and individual technical questions.	HW 6 due Dec. 10; presentation groups assigned in Canvas.
15Dec. 15	Team presentations, individual questions, lessons from inspection/damage indicators, and course synthesis	Final project package due 5:00 p.m.

Homework map

Assignment	Primary focus	Due
HW 1	Read the structure: airfoil/wing load path, FBD, pre-calculation predictions, equilibrium, code plot	Oct. 1
HW 2	HW 2A: Lilienthal evidence and side-force geometry (35 points).	Oct. 20, 2026
HW 3	Three distributed-load beams: reactions, compatibility and selected displacement/rotation, analytically and with software (65 points).	Nov. 5, 2026
HW 4	Audit AI reasoning: virtual work/Castigliano plus correction of a plausible but flawed generated solution	Nov. 19
HW 5	Sensitivity and trade-offs: maneuver/landing loads, V-n diagram, computational exercise	Dec. 3
HW 6	Integrated trust case: wing-box shear/torsion/shear center; hand-code/AI validation; design/manufacturing evidence; calibrated vision inspection; and interpretation of shared damage-diagnostic data	Dec. 10

8. Course policies

Attendance, participation, and make-up work

I do not perform presence check every class, but regular attendance is expected because studios, oral checks, design reviews, speaker events, and safety training cannot be reconstructed fully from slides. Notify the instructor as early as practicable for illness, emergency, military duty, religious observance, disability-related needs, or another University-recognized absence. Approved absences receive a reasonable equivalent activity or make-up without penalty. Students remain responsible for announcements and for coordinating team handoffs.

Deadlines and extensions

Each student may use two 24-hour homework extensions during the semester without giving a reason. The passes do not apply to the midterm, oral rotation, project safety gates, printer reservations, fabrication/assembly appointments, or the final presentation. Beyond the passes, late individual work loses 10% of the earned score per 24 hours unless an approved circumstance or accommodation applies. Contact the instructor before a project deadline whenever possible; team safety and printer capacity may require an alternate deliverable rather than a shifted machine slot.

Religious observance

Students are entitled to reasonable accommodation for religious observances. Review the schedule early and notify the instructor so an alternative can be arranged without penalty. See the UMass Academic Regulations.

Academic integrity and collaboration

Engineering depends on trustworthy evidence. Discussion of concepts and comparison of approaches are encouraged unless an assignment says otherwise, but every individual homework and midterm submission must represent the student's own reasoning. Project teammates may share team artifacts; individual contribution records and oral questions remain individual. Copying prior solutions, submitting another person's code or text, fabricating analysis, process records, or citations, or using unauthorized assistance is prohibited.

The University Academic Integrity Policy applies to all work: UMass Academic Integrity Policy. Suspected violations follow University procedures.

AI Statement: Learning and Engineering Judgment in the Age of AI

Generative AI can produce equations, code, plots, and polished explanations in seconds. This changes what counts as evidence of learning, but it does not transfer the engineer's responsibility. The course is organized around a central question: When AI can produce a solution, what must an aerospace engineer still understand and be able to do? Students must frame the problem, predict behavior, choose and justify the model, test the result against mechanics and evidence, recognize uncertainty and limitations, and defend the decision.

This course uses a default-permitted, evidence-required approach unless an assignment restricts a component. AI may support learning, brainstorming, alternative approaches, debugging, communication, or adversarial review. AI output is material to inspect; it is not evidence that a claim is true. Treat it as an unverified assistant: potentially useful, never self-validating. Students remain responsible for safety, evidence, attribution, empathy, professional judgment, teamwork, and the ability to reproduce and defend the essential reasoning.

Rule	Meaning in this course
Permitted with disclosure	Learning, brainstorming, examples, alternative approaches, code debugging, language revision, and adversarial critique - when the assignment does not restrict the tool
Required evidence	Tool and purpose; material prompt/output excerpts when requested; high-impact claims used; student verdict on technical AI output—Accept, Accept with limitations, or Reject; changes and errors found; verification by mechanics, authoritative sources, independent calculation, code tests, slicer evidence, dimensional checks, or documented fabrication evidence; known limitations
Not permitted	Midterm or a restricted component; undisclosed generated work; fabrication of citations/data; hiding lack of understanding; replacing a teammate's responsibility; uploading restricted course or teammate material
Accountability	The student owns every equation, number, sentence, citation, code line, and decision, and must be able to explain the model, reproduce essential reasoning, identify limitations, predict a controlled change, and defend the work

AI detectors will not be used as proof of misconduct. Evidence of learning comes from transparent reasoning, prediction records, independent checks, reproducible code, process history when requested, physical fabrication evidence, revision after error, one-slide synthesis, and the student's ability to explain and defend the work. When uncertain, ask before submitting.

Regrades

Submit a concise written regrade request within two calendar days of the grade posting. Identify the specific rubric item and evidence; do not ask only for additional points. The complete submission may be reviewed, and the score may increase, remain unchanged, or decrease if another material grading error is found.

Recording, privacy, and course materials

Do not record classmates, guest speakers, oral checks, fabrication sessions, or project reviews without permission. Course files may be used for enrolled students' learning but may not be publicly posted, sold, or uploaded to answer-sharing or AI-training services. Students should remove private identifiers from shared project data.

9. University support and required statements

Disability accommodations

UMass Amherst is committed to equal educational access. Students who need an accommodation should connect with Disability Services and provide the course accommodation notice as early as possible. Accommodations apply to classrooms, assessments, oral alternatives, and project/lab access. Disability Services.

Required University statements and Title IX resources

University policies regarding accommodations, academic integrity, and Title IX apply to this course. In accordance with current Faculty Senate guidance, this syllabus uses the required-statement route for a non-responsible employee. Confidential and non-confidential support resources remain available whether or not a student makes a formal report.

Required statements: UMass Faculty Senate - Non-Responsible Employee Required Syllabus Statements. This page contains the current Academic Integrity, Accommodation, and Title IX statements incorporated into this syllabus by reference.

Health and wellbeing

Coursework and life demands can become difficult. Contact the instructor early when circumstances affect participation, and use campus support. Confidential mental-health services are available through the Center for Counseling and Psychological Health. In an immediate emergency, call 911 or use the University’s emergency resources.

Inclusive learning environment

Aerospace engineering benefits from different experiences, identities, and approaches to problem solving. Everyone is expected to use classmates’ names and pronouns, critique ideas rather than people, share technical work equitably, and support a classroom and lab free of discrimination, harassment, and retaliation. Students may raise a concern privately with the instructor or through the appropriate University office.

Safety and stop-work authority

Every student has authority to pause a print, tool use, or assembly when something appears unsafe. Stop work, keep others clear, and notify the instructor/TA/lab staff. Safety reporting will never reduce a project grade; bypassing required controls may.

Official policy sources

- UMass syllabus requirements
- Academic Integrity Policy
- Responsible use of generative AI
- Courses Extra Credit Policy
- Registrar Academic Calendar